

SHAKYA MARASINGHA

Materials Science & Engineering Undergraduate

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[linkedin.com/in/shakya-marasingha-5aa11426b](https://www.linkedin.com/in/shakya-marasingha-5aa11426b) | [Shakya Marasingha](#) | [Materials Science Engineer](#)

PROFESSIONAL SUMMARY

Third year Materials Science & Engineering undergraduate at the University of Moratuwa with strong interest in advanced materials, nanotechnology, electrochemical systems, energy storage, and sustainable engineering. Experienced in interdisciplinary research involving mathematical modelling, computational analysis, reaction kinetics, electrochemistry, and quantum-inspired materials science. Currently building skills in machine learning and MATLAB to complement engineering and research work. A proactive team player with leadership experience in IEEE and university organizations, seeking internship and research opportunities to further develop technical and professional expertise.

EDUCATION

Bachelor of the Science of Engineering Honours (B.Sc. Hons Eng.) — Materials Science & Engineering | University of Moratuwa | *March 2024 – Present*

- Relevant coursework: Materials Characterization, Nanomaterials, Electro-materials, Electrochemical Systems, Materials Thermodynamics, Materials Kinetics, Functional Materials, Sustainable Engineering

Diploma in Information Technology | ESOF Metro Campus | *April 2023 – May 2024*

Aquinas Diploma in English Language and Literature | BCI Campus | *April 2023 – December 2023*

WORK EXPERIENCE

General / Operations Intern | Bank of Ceylon (BOC) | *2023 – 2024*

- Provided customer service support, assisting clients with routine banking queries and transactions
- Supported marketing initiatives to promote bank products and services
- Assisted with account handling and account-related administrative tasks
- Supported front-office and back-office staff with financial transactions and record-keeping
- Developed practical exposure to the banking and financial services sector, strengthening attention to detail and professionalism

PUBLICATIONS

“Are Solar Cells Really Eco-Friendly?” | Article | *2026*

- Examined the environmental impacts of solar cells across their full life cycle — from raw material extraction and manufacturing to recycling and disposal
- Applied a life cycle assessment perspective to evaluate the sustainability of solar energy technology
- Concluded that solar energy remains one of the most sustainable solutions despite environmental challenges associated with its production

Read the article: <https://lnkd.in/g7eQjHtX> or <https://lnkd.in/gcwYQ2DH>

ENGINEERING PROJECTS

Artificial Photosynthesis: From Biological Blueprint to Quantum Engineering | University of Moratuwa — MT2181, Solid State Materials

- Investigated quantum mechanical principles governing artificial photosynthesis
- Studied semiconductor photocatalysts and electronic band structures
- Analysed charge separation, electron transport, and photocatalytic reactions
- Explored Z-scheme mechanisms, water oxidation, hydrogen evolution, and quantum efficiency
- Integrated concepts from quantum mechanics, materials science, electrochemistry, and renewable energy

Kinetic Modelling of Dopamine Transport in the Synaptic Cleft | University of Moratuwa — MT2230, Kinetics of Materials

- Developed a mathematical model describing dopamine transport using Poisson–Nernst–Planck equations
- Incorporated diffusion, electro-diffusion, Michaelis–Menten kinetics, and receptor binding models
- Investigated physiological and pathological neurotransmitter transport
- Applied computational and analytical methods to analyse dopamine concentration profiles

LEAP Laptop Stand | Engineering Design & Workshop Project

- Designed and developed an ergonomic laptop stand prototype with adjustable design, cooling-perforated wooden platform, natural tea-waste thermal insulation, and memory-foam cushioning
- Applied sustainable and user-centred design principles throughout

Innovative Seed Spreader — Machine Element Design | University of Moratuwa — ME2851, Fundamentals of Machine Elements Design

- Conducted market survey and analysed existing seed spreader designs, identifying limitations in seed distribution, control, and portability

- Proposed an improved design featuring an adjustable smart seed metering system, dual distribution discs, variable gear ratio transmission, and a seed level indicator
- Performed engineering calculations for gear ratio, shaft diameter, bearing selection, and power requirements using machine design principles
- Produced conceptual drawings and conducted a SWOT analysis to evaluate design feasibility and market potential

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Team Lead, Publicity Committee (Octwave 3.0) | IEEE Student Branch, University of Moratuwa | *April 2026 – Present*

- Lead publicity activities and coordinate promotional campaigns
- Manage team collaboration and event communications

Team Member, Publicity Committee (Moraforesight 4.0) | IEEE Student Branch, University of Moratuwa | *April 2026*

Chairperson, Field Visit | IEEE NPSS Student Branch Chapter | *September 2026 – October 2026*

Team Member, Publicity Committee | IEEE NPSS Student Branch Chapter | *April 2026 – Present*

Co-Chair, Energy Market Webinar Session | IEEE PES Student Branch Chapter | *March 2026 – April 2026*

- Coordinated webinar planning and execution
- Assisted with speaker communication and event management

Team Member, Organizing Committee | IEEE PES Student Branch Chapter | *February 2026 – Present*

Marketing Team Member & Finance Panel Member | Mora Lenz | *February 2026 – Present*

- Hosted the team for year-long sponsorship programme

Team Member, Editorial Committee | SOMES | *July 2026 – Present*

Student Member | Institution of Engineers – Sri Lanka | *July 2024 – Present*

- Currently holding the membership in IESL – membership no. S33302

Committee Member | Classical Music Society, University of Moratuwa | *October 2025 – Present*

VOLUNTEERING

Team Member, Arts & Culture | Rotaract Club of University of Moratuwa | *April 2025 – May 2025*

- Participated in community engagement activities while developing teamwork and leadership skills

EXTRACURRICULAR ACTIVITIES

Team Player | University of Moratuwa Hockey Team

- Represent the university at inter-university hockey tournaments as part of the team
- Developed discipline, teamwork, and time-management skills through regular training and competitive play

TECHNICAL SKILLS

Engineering: Materials Science, Materials Characterization, Nanomaterials, Electrochemical Systems, Functional Materials, Sustainable Engineering, Materials Kinetics, Solid-State Physics

Research: Scientific Literature Review, Mathematical Modelling, Computational Analysis, Data Interpretation, Technical Report Writing

Software & Programming: AutoCAD, Solid-Edge, ImageJ, Python, C#, MATLAB (Academic, in progress), Machine Learning (in progress), Microsoft Excel, Microsoft Word, Microsoft PowerPoint

Professional Skills: Analytical Thinking, Research Skills, Leadership, Teamwork, Problem Solving, Project Coordination, Communication, Time Management

PROFESSIONAL AFFILIATIONS

- IEEE Student Branch – University of Moratuwa
- IEEE Nuclear & Plasma Sciences Society (NPSS)
- IEEE Power & Energy Society (PES)
- Rotaract Club of University of Moratuwa
- SOMES
- Classical Music Society – University of Moratuwa
- AIESEC CS

REFERENCES

- Prof. Dinesh Attygalle
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